

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCVSS801F - ENGINEERING SYSTEMS & DEVELOPMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 What is an example of a systems engineering product?
 - a) A standalone software tool
 - b) A complete transportation system design
 - c) A single mechanical component
 - d) A marketing campaign
- 2W Why is development testing conducted in the engineering development stage?
 - a) To finalize the system's operational phase
 - b) To validate system performance and reduce risks
 - c) To design the system's user interface
 - d) To conduct maintenance operations
- 3 Development in engineering mainly aims to:
 - A. Increase cost
 - B. Improve efficiency and performance
 - C. Reduce manpower only
 - D. Stop innovation
- 4W What is a critical aspect of upgrades and modernization in the production operations phase?
 - a) Developing initial system requirements
 - b) Enhancing system performance to meet evolving needs
 - c) Designing the system architecture
 - d) Conducting concept exploration
- 5W What is the primary goal of trade-off analysis in systems engineering?
 - a) To validate system requirements
 - b) To balance competing objectives and optimize solutions
 - c) To finalize the system architecture
 - d) To allocate system functions
- 6 Redundancy in system design is used to achieve which goal?
 - a) Increased reliability
 - b) Simplified architecture
 - c) Cost reduction
 - d) Enhanced maintainability

- 7 Which of the following is a renewable energy resource used in engineering systems?
 - A. Coal
 - B. Petroleum
 - C. Solar energy
 - D. Natural gas
- 8 Which activity aims to mitigate program risks during development?
 - a) Prototype development
 - b) Final production testing
 - c) User interface design
 - d) Installation and upgrades
- 9 What is the role of a Work Breakdown Structure (WBS) in systems engineering?
 - a) To design system components
 - b) To break down complex projects into manageable tasks
 - c) To eliminate all system risks
 - d) To conduct final system testing
- 10 Which statistical method is used to test hypotheses in systems development?
 - a) Linear regression
 - b) Hypothesis testing
 - c) Data interpolation
 - d) Correlation analysis
- 11 In a complex system structure, what does the term "hierarchy" refer to?
 - a) The sequence of production stages
 - b) The organization of subsystems in a layered structure
 - c) The external environment of the system
 - d) The elimination of system interfaces
- 12 What is a key characteristic of the system development process lifecycle?
 - a) It is static and does not evolve
 - b) It involves iterative phases and feedback loops
 - c) It excludes stakeholder involvement
 - d) It focuses only on system disposal

Q2. Solve the following.

- A) Explain the concept of systems engineering and provide two real-world examples of its application in modern industries 6
- B) Describe the different domains and perspectives of systems engineering and how they contribute to system development. 6

Q3. Solve the following.

- A) Outline the main fields within systems engineering. Discuss the approaches and activities typically involved in each field, and provide examples of the types of products that result from these activities. 6
- B) Describe the systems development process and its life cycle stages. 6

Q4. Solve Any Two of the following.

- A) What is a Work Breakdown Structure (WBS), and how is it applied in systems engineering? 6
- B) Explain the importance of change management in engineering design. How can effective change management practices mitigate the risks associated with design changes? 6
- C) What is functional analysis and allocation? Explain its role in concept definition. 6

Q5. Solve Any Two of the following.

- A) Explain the significance of hypothesis testing in system development. 6

- B) Explain the process of system integration and its challenges in ensuring total system functionality. 6
 - C) Describe the role of in-service support in maintaining the functionality and reliability of deployed systems. How do you prioritize and allocate resources for in-service support activities? 6
- Q6. Solve Any Two of the following.
- A) Describe the importance of integration testing in the software development lifecycle. How does it ensure the reliability and functionality of the entire system? 6
 - B) Explain the role of user interface (UI) design in systems engineering. 6
 - C) Discuss the importance of in-service support in maintaining system performance and reliability. 6

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